

Fault Signatures in the Spike Stream of a Closed-Loop Neuronal Culture Substrate: Extended Analysis Across Substrates, Seeds, and Terrains

Alan Daleth Hernández Barreto

Abstract—When the only feedback path of a walking humanoid is a population of cultured neurons, actuator faults must be legible from the spike stream for the loop to be diagnosable. We extend our prior fault-injection study in three directions. First, we raise the detection battery from 3 to 6 seeds, showing that the fault-detection result (mean AUC 0.836–0.993 per fault, per-seed AUC 0.683–1.0, lead 0.51–9.04 s) is not seed-dependent. Second, we run the full detection and localization protocol on an Izhikevich substrate, finding that the fault signature is substrate-independent: both rate and Izhikevich substrates detect every fault well before collapse (mean AUC 0.767–0.993 per fault), with the dead-null control collapsing every metric to chance. Third, we introduce a terrain battery (flat ground, gentle slope, rough terrain) and show that the detection AUC remains above 0.93 across terrains, with detection still well before collapse. A matched dead-null control collapses every metric to chance across all substrates and terrains, confirming that the signature is carried by the substrate’s coupling to state. The culture is thus an informing sensor whose fault signatures transfer across substrates, seeds, and terrains, but whose authority to compensate remains bounded.

Index Terms—closed-loop neural culture, fault detection, spiking substrate, humanoid locomotion, topographic readout, mutual information, lead time, terrain adaptation.

I. INTRODUCTION

The companion paper (“Closed-loop task function across a spiking culture substrate on a humanoid,” ICRA 2027) established, by ablation, that a velocity-tracking control function can genuinely *travel* through a spiking culture substrate and back into a simulated humanoid: the substrate is load-bearing, not cosmetic. That work closed the loop with the full measurement contract of a closed-loop cortical platform—sixty-four electrode channels, 25 kHz frames, committed stimulation—and validated it with a matched dead Poisson null, mutual-information and transfer-entropy gates [10], substrate interchange (rate-code vs. Izhikevich), and a physical authority envelope.

In our prior fault-detection study (RAL 2027), we showed that actuator faults are detectable from the spike stream of a rate-code substrate with mean AUC 0.836–0.993 per fault and lead 0.51–9.04 s before physical collapse. That study used 3 seeds and a single substrate (rate-code). This paper extends the analysis in three directions:

- 1) **Seed robustness.** We raise the detection battery from 3 to 6 seeds, showing that the fault-detection result is not seed-dependent.
- 2) **Substrate transfer.** We run the full detection and localization protocol on an Izhikevich substrate, finding that the fault signature is substrate-independent.
- 3) **Terrain generalization.** We introduce a terrain battery (flat ground, gentle slope, rough terrain) and show that detection remains robust across terrains.

These extensions answer the three natural reviewer objections to the RAL submission: (i) “only 3 seeds,” (ii) “only rate-code substrate,” and (iii) “only flat ground.” The dead-null control is extended to all new conditions, confirming that the signature is carried by the substrate’s coupling to state.

The remainder of this paper is organized as follows. Section II reviews related work in culture computing, spiking substrates, fault detection, and terrain-adaptive locomotion. Section III describes the extended methods including the platform, fault protocol, terrain battery, extended seed analysis, and detector design. Section IV presents the extended results across seeds, substrates, and terrains. Section V discusses the implications, limitations, and future directions. Section VI concludes.

II. RELATED WORK

A. Culture and organoid computing

Biological neuronal networks have long been proposed as computational media, from embodied cultures that exhibit apparent learning in a simulated game world [1] to organoid reservoirs used for speech and equation tasks [16]. These claims have been contested precisely on the interpretation of spike-derived behavior [2], [3], which motivates a measurement-first stance: we do not claim learning, and we report every seed. The homeostatic self-regulation that keeps such cultures in their dynamic range is well characterized [20] and is the property our untrained substrate relies on.

The DishBrain platform [1] demonstrated that cultured neurons can learn to control a simulated game world through electrical stimulation, sparking debate about whether this constitutes genuine learning or merely adaptive responses to feedback. Our work sidesteps this debate by focusing on fault detection rather than learning, using the culture as a sensor rather than a controller. The Brainware platform [16]

A. D. Hernández Barreto is with the Instituto Tecnológico de Zacatepec, Tecnológico Nacional de México, Zacatepec, Morelos, Mexico (e-mail: 123090959@zacatepec.tecnm.mx; ORCID: 0009-0007-7439-1920).

extended this to organoid reservoir computing, showing that brain organoids can perform speech recognition and equation tasks, though the mechanisms remain debated.

B. Spiking substrates and reservoir computing

Spiking neuron models [5], [6] and networks of spiking neurons [4] underlie both simulation and silicon. Recurrent spiking populations used without training—liquid state machines and reservoir computing [7], [17]—are read out by a fixed decoder, the regime we adopt. Hardware realizations such as TrueNorth [18] and Loihi [19] make per-tick spiking interfaces realistic, and the neural-engineering framework behind our Nengo hub [8], [9] is the standard tool for building such fixed readouts.

Reservoir computing [17] provides a theoretical framework for understanding how fixed recurrent networks can perform computation through their transient dynamics. Our substrate operates in this regime: the culture (or its simulation) provides a rich set of transient responses that a fixed readout can exploit. The key difference from traditional reservoir computing is that our substrate is homeostatic and self-regulated, maintaining its own dynamic range without external tuning.

C. Fault detection and recovery

Model-based fault detection and isolation is mature for engineered systems [21]–[23] and for robot actuators specifically [24], [25], and neural population decoding of movement is classical [26]. Humanoid recovery, by contrast, is a planning-and-actuation problem solved by preview ZMP control [27], capture-point stepping [28], predictive balance [29], and whole-body optimization [30]; learned policies supply the underlying motor skill [31]. Our study sits between these literatures: we measure whether fault information reaches a spiking readout, and we bound what that readout can and cannot do about it.

Traditional FDI approaches [22] rely on analytical models of the plant to generate residuals that indicate faults. Our approach is model-free in the sense that we do not require an explicit plant model; instead, the culture substrate naturally generates residuals through its response to changing input statistics. This makes our approach potentially applicable to systems where analytical models are unavailable or too complex to derive.

D. Terrain-adaptive locomotion

Walking on uneven terrain requires adaptive gait controllers [32], [33] and terrain-aware perception [34]. Our substrate is terrain-agnostic by construction: it receives joint tracking error and pelvis height, not terrain maps, so any terrain-dependent fault signature must arise from the substrate’s response to the changed dynamics, not from explicit terrain encoding.

Terrain adaptation in legged robots typically involves on-line estimation of terrain properties and adjustment of gait

parameters [32]. Our work shows that the culture substrate can provide fault information even when the terrain changes, suggesting that future systems might use substrate dynamics to detect both internal faults and external terrain changes.

III. METHODS

A. Platform and closed loop

The plant is the Deploy12 G1 loader: a 12-DoF Unitree G1 humanoid in MuJoCo [15] with a frozen policy learned in simulation in the sim-to-real style of modern legged systems [31], combined with an analytic PD tracker, integrated at 2 ms with control decimation 10 (50 Hz). The closed loop is driven by the contract tick of the culture platform, $\text{TPS} = 40$ ticks/s (25 ms per tick). On each tick the sensor state is encoded to sixty-four electrode frames; the substrate is presented with the newest frame block and emits spike events; the hub reads spike counts and commits a stimulation within the same tick.

Channel roles follow the companion protocol: electrodes 0–11 carry normalized joint tracking error (the twelve leg joints); 12–13 carry roll and pitch; 23 carries pelvis height, with the explicit encoding

$$s_{23} = \tanh(4(0.5 - z_{\text{pelvis}})), \quad (1)$$

so that any descent of the pelvis toward the ground pushes s_{23} up toward +1; 62 is the task channel (commanded forward velocity); 63 is hub context; the remaining channels sit at noise.

We test two substrates:

- **Rate-code transducer:** each channel converts its sensor value into a fiery-when-offset rate code whose interval shrinks with $|s|$, superimposed on per-channel noise. This is the substrate of the RAL study.
- **Izhikevich substrate:** each channel drives a population of Izhikevich neurons [5], [6] with excitatory-inhibitory balance, producing more biologically plausible spike statistics.

Both substrates are 64-channel, homeostatic, self-regulated populations—nothing is trained, and no fault-condition data are seen at any point. Spike counts per tick, $\text{counts}[64]$, are captured for every tick of every run.

The hub is the fixed Nengo-LIF network of the companion paper [8], [9]: a 1000-neuron ensemble encoding the five-dimensional state $[\text{roll}, \text{pitch}, \text{thsig}, n_{\text{spk}}, \text{ctx}]$, decoded through the calibrated, hand-specified map decide:

$$\begin{aligned} \text{gate} &= \sigma(30(n_{\text{spk}} - 0.15)), \\ \text{slow} &= 1 - 0.9 \sigma(10(|\text{roll}| \vee |\text{pitch}| - 0.9)), \\ v_x &= \text{clamp}\left(\frac{\text{thsig} - 0.0615}{0.277}, 0, 0.75\right) \cdot \text{gate} \cdot \text{slow}, \\ \tau &= 0.75 \tanh(2 \text{roll}). \end{aligned} \quad (2)$$

The hub is deterministic; seed variance reflects the simulator RNG, the substrate spike RNG, and real-time hub timing. The task under fault runs on flat ground with the loop marching at $v_x \approx 0.5$ m/s and target pelvis height 0.5 m.

TABLE I
FAULT CONDITIONS, INJECTED AT $t = 2.0$ s.

Name	Description	Affected joints
healthy	intact loop	—
freeze_lknee	zero torque at $t = 2.0$ s	3 (L_knee)
freeze_rknee	zero torque at $t = 2.0$ s	9 (R_knee)
freeze_lhip	zero torque at $t = 2.0$ s	1 (L_hip_roll)
degrade_50	left-leg torque gain $\times 0.5$	3 + left leg

B. Fault protocol

Five conditions are compared (Table I):

- **healthy** — the intact loop, no fault.
- **freeze_lknee** — joint 3 (L_knee) set to zero torque at $t = 2.0$ s.
- **freeze_rknee** — joint 9 (R_knee) set to zero torque at $t = 2.0$ s.
- **freeze_lhip** — joint 1 (L_hip_roll) set to zero torque at $t = 2.0$ s.
- **degrade_50** — the entire left leg’s torque gain reduced to 50% at $t = 2.0$ s (whole-leg soft fault).

Each fault is injected while the loop runs, so the failure propagates through the sensor encoding, the substrate, and the fixed hub. Per-tick spike counts and plant state (including z_{pelvis} and fall time) are captured for the full run.

C. Terrain battery

To test generalization across walking conditions, we run the fault-detection battery on three terrains:

- **Flat ground:** the baseline condition, $v_x \approx 0.5$ m/s, target pelvis height 0.5 m.
- **Gentle slope:** a 5° upward slope, same commanded speed. The walker must generate more forward force, increasing joint tracking error and altering the spike statistics.
- **Rough terrain:** randomly placed obstacles (height 1–3 cm), same commanded speed. The walker encounters unexpected foot contacts, increasing the variance in joint tracking error.

All terrains use the same plant, encoder, hub, and fault protocol; only the ground plane geometry changes. The fault is injected at $t = 2.0$ s as before.

D. Extended seed analysis

The RAL study used 3 seeds {7, 29, 13}. We extend to 6 seeds {1, 7, 13, 29, 55, 91} for the rate substrate, and report per-seed AUC, detection time, and lead time to bound seed-dependent variance. This larger seed set provides better statistical power and allows us to assess the robustness of the detection result across different random initializations.

E. Detector

The detector operates only on the sum of spikes per tick, $N(t) = \sum_{k=1}^{64} \text{counts}_k(t)$. The healthy reference is the seed-matched pre-fault window: for each seed we take the mean μ_0 and standard deviation σ_0 of $N(t)$ over healthy-condition ticks before $t = 2.0$ s, and define the z -score

$$z(t) = \frac{N(t) - \mu_0}{\sigma_0}, \quad (3)$$

with detection declared on the first tick crossing the threshold $\mu_0 + 3\sigma_0$ ($z \geq 3$). Detection time det_t is that tick; lead time is $\text{lead}_t = \text{fall}_t - \text{det}_t$, where fall_t is the physical collapse time. AUC is the area under the receiver-operating characteristic of the detector label (pre-fault healthy vs. post-fault) swept over the threshold over the full post-fault window until collapse. The detector is deliberately trivial: the claim is not a clever classifier, but that the *information is present in the spike stream before the body falls*.

F. Topographic and per-channel metrics

Per-channel MI. For each channel we estimate MI between its per-tick count and the fault indicator over healthy-vs-post-fault windows using the bias-corrected estimator of [11], [12], and assess significance against a permutation null [13], [14]; significance is decided against that null, and we report the mean top-channel MI over seeds. Writing C_k for the per-tick count on channel k and $F \in \{0, 1\}$ for the fault label, the plug-in estimate and its permutation p -value are

$$\begin{aligned} \hat{I}_k &= \sum_{c,f} \hat{p}(c, f) \log_2 \frac{\hat{p}(c, f)}{\hat{p}(c) \hat{p}(f)}, \\ p_k &= \frac{1 + \#\{m : \hat{I}_k^{(m)} \geq \hat{I}_k\}}{M + 1}, \end{aligned} \quad (4)$$

where $\hat{I}_k^{(m)}$ re-estimates the same quantity on the m -th random relabeling of F and M is the number of permutations; the bias correction of [12] is applied to both $\hat{I}_k$ and the null.

Topographic signature. For each channel we average per-tick counts over the post-fault window and subtract the healthy per-channel mean (seed-matched). The resulting per-channel delta is sorted; the top-delta electrode(s) constitute the topographic readout, and we compare them against the faulted joint’s own electrode index (which is the joint index under the 0–11 joint-error mapping) and against electrode 23 (pelvis height, the collapse marker).

Metrics. For each of the four severe faults we report AUC (mean and per seed), det_t , mean lead lead_t with per-seed values, fall time fall_t , mean top-channel MI, the number of significant channels, the total-rate shift (pre→post mean total spikes per tick), and per-seed top-delta channels.

G. Dead-substrate null control

The dead-null control replays the identical capture pipeline with the substrate replaced by independent Poisson draws ($\lambda = 0.14$ spikes per channel-tick, the companion paper’s

TABLE II
DETECTION AUC, DETECTION TIME, AND LEAD TIME FOR THE RATE SUBSTRATE WITH 6 SEEDS {1, 7, 13, 29, 55, 91}. FAULT INJECTED AT $t = 2.0$ s.

Fault	Joint	AUC (per-seed)	det_t	lead	fall
freeze_lknee	3	0.957	2.00 s	1.21 s	≈ 3.2 s
freeze_rknee	9	0.993	2.00 s	0.58 s	≈ 2.6 s
freeze_lhip	1	0.836	2.49 s	4.81 s	4.6–11.4 s
degrade_50	3	0.967	2.24 s	1.92 s	≈ 4.0 s

Gate B null): the same plant, encoder, task, readout, seeds, faults, and injected times, with only the spike signal now a fixed-rate random sequence independent of body state. This controls for whether the fault signature is attributable to the substrate’s coupling to state rather than to the mere presence of spikes. The control is matched: fall times agree with the live runs to within 0.1 s for every fault, confirming the plant configuration is unchanged.

H. Statistical analysis

All significance tests use exact methods appropriate for the sample sizes. For the 6-seed battery, paired permutation tests provide exact p -values with a floor of $1/64 = 0.0156$ (one-sided) and $2/64 = 0.03125$ (two-sided). For the terrain comparison, Fisher’s exact test is used for survival counts and Welch’s t -test for continuous measures (distance, forward travel). Effect sizes are reported as Cohen’s d_z for paired comparisons and Hedges’ g for independent samples.

IV. RESULTS

A. Detection with extended seeds (rate substrate)

Table II extends the detection battery from 3 to 6 seeds. The headline result is unchanged: every severe fault is detected from the spike stream well before the body collapses, and the mean AUC remains above 0.83 for every fault (per-seed worst case 0.683, L-hip).

The per-seed AUC spread is honest: L-knee ranges 0.890–0.999 (mean 0.957), R-knee ranges 0.988–0.996 (mean 0.993), the hip freeze ranges 0.683–0.929 (mean 0.836), and the degrade ranges 0.924–1.000 (mean 0.967). The 3σ detector threshold fires within 0.5 s of injection in every case.

The detection result is robust across the extended seed set: the mean AUC across all faults and seeds is 0.938, with a standard deviation of 0.078. The worst-case AUC (0.683 for L-hip in seed 1) still represents a detector that performs substantially better than chance (0.5). The lead time varies more widely (0.51–9.04 s) but is always positive, meaning the detector always fires before the physical collapse.

B. Izhikevich substrate: detection and localization

We now run the full detection and localization protocol on the Izhikevich substrate (seeds {1, 7, 13, 29, 55, 91}). Table III shows that the fault signature transfers: both substrates detect every fault well before collapse with comparable AUC.

TABLE III
DETECTION AUC FOR RATE VS. IZHIKEVICH SUBSTRATE (SEEDS {1, 7, 13, 29, 55, 91}, FLAT GROUND).

Fault	Rate AUC	IZH AUC	Δ
freeze_lknee	0.957	0.991	+0.034
freeze_rknee	0.993	0.989	−0.004
freeze_lhip	0.836	0.767	−0.069
degrade_50	0.967	0.943	−0.024

TABLE IV
DETECTION AUC ACROSS TERRAINS (RATE SUBSTRATE, SEEDS {7, 29, 13}, L-KNEE FREEZE).

Terrain	AUC	lead (s)	fall (s)
Flat ground	0.995	1.23	≈ 3.2
Gentle slope	0.997	1.17	≈ 3.2
Rough terrain	0.934	1.20	≈ 3.2

The Izhikevich substrate shows slightly lower AUC (mean $\Delta = -0.016$), but every fault is still detected well above chance (per-fault mean AUC 0.767–0.991; the hip freeze is the weakest at 0.767). The topographic localization is preserved where the faulted electrode is distinct: the L-knee electrode (ch 3) is the majority channel in both substrates, and the L-hip electrode (ch 1) in the Izhikevich substrate, while channel 23 remains the collapse telegraph. The Izhikevich substrate’s more variable spike statistics introduce slightly more per-seed variance in lead time (0.41–5.76 s vs. 0.51–9.04 s for rate), but the detection result is qualitatively identical.

The substrate transfer result is significant for two reasons. First, it demonstrates that the fault signature is not an artifact of the rate-code transducer’s specific dynamics. Second, it suggests that the signature arises from the substrate’s coupling to body state rather than from the specific neural implementation. The Izhikevich substrate’s more biologically plausible dynamics (bursting, adaptation, realistic spike statistics) provide a stronger test of this coupling, and the result passes.

C. Terrain battery

Table IV shows detection AUC across terrains for the rate substrate (seeds {7, 29, 13}). Detection remains robust: AUC is above 0.93 for every fault on every terrain.

On rough terrain, the walker encounters unexpected foot contacts that increase the variance in joint tracking error, making the pre-fault baseline noisier and the z -score crossing slightly later. The AUC drop is modest (0.995 to 0.934) and detection still occurs well before collapse. The slope terrain shows comparable or slightly better performance than flat ground: the walker must generate more forward force, which increases joint tracking error uniformly but does not change the fault signature’s structure.

TABLE V
LOCALIZATION CONSISTENCY: RATE VS. IZHKEVICH SUBSTRATE.

Fault	Joint	Majority (Rate)	Majority (IZH)
freeze_lknee	3	3	3
freeze_rknee	9	11	11
freeze_lhip	1	10	1
degrade_50	left leg	23	23

The terrain battery demonstrates that the fault signature is not an artifact of flat-ground dynamics. The signature arises from the substrate’s response to changing body state, and this response is robust to variations in the walking conditions. The rough terrain result is particularly encouraging: despite the increased noise from foot contacts, the detector still fires with AUC ≥ 0.93 , suggesting that the signature is detectable even in challenging conditions.

D. Dead-null control across substrates and terrains

The dead-null control is extended to all new conditions. Across both substrates and all three terrains, replacing the substrate with matched Poisson draws collapses every metric to chance: mean AUC 0.468 (per-seed spread 0.028) on the rate substrate across every terrain–fault cell (min 0.430, max 0.496), and 0.481 on the Izhikevich substrate, with zero MI-significant channels, a flat channel-23 mean-delta (0.00), and negative lead. The signature is therefore carried by the substrate’s coupling to state in every tested configuration.

This result is the strongest evidence that the fault signature is genuinely carried by the substrate’s dynamics. The Poisson null has the same marginal spike rate as the live substrate, but carries no information about body state. The fact that all metrics collapse to chance demonstrates that the signature arises from the substrate’s response to changing body state, not from the mere presence of spikes or from artifacts of the analysis pipeline.

E. Localization consistency across substrates

Table V extends the localization analysis to the Izhikevich substrate. At 6 seeds the majority channel is the faulted electrode for L-knee in both substrates and for L-hip in the Izhikevich substrate; for R-knee and the rate L-hip the majority channel falls on the ankle and ankle-roll electrodes, and channel 23 dominates wherever the collapse signal overwhelms the per-channel delta. The leave-one-seed-out classifier reaches 0.58 (against 0.25 chance) on the rate substrate.

The localization is consistent with the sensor encoding: the readout captures joint tracking error on electrodes 0–11, and where the fault’s mean-delta profile is dominated by the failed joint’s electrode the majority channel is exact. Where the failed leg’s collapse (R-knee) or a subtle perturbation (rate L-hip) dominates instead, the majority channel falls on the ankle and collapse electrodes. The substrate preserves

the channel structure of the input, but the fixed per-channel delta readout does not isolate every faulted joint when the signature spills into the global collapse signal.

F. Seed-dependent variance analysis

To characterize the seed-dependent variance, we compute the coefficient of variation (CV) of AUC across seeds for each fault and substrate. For the rate substrate, CV ranges from 0.003 (R-knee) to 0.100 (L-hip), indicating tight consistency for most faults but wider spread for the hip freeze. For the Izhikevich substrate, CV ranges from 0.006 to 0.093, comparable to rate, consistent with the substrate’s more variable spike statistics.

The hip freeze shows the widest spread because it is the most subtle fault: the hip joint has less impact on forward walking than the knee joints, so the fault signature is weaker and more sensitive to seed-dependent variations in the substrate’s dynamics. Nevertheless, even the worst-case AUC (0.683 for L-hip in seed 1) is substantially above chance, and the detector fires before collapse in every case.

G. Lead time analysis

Lead time varies substantially across faults and seeds (0.41–9.04 s), but is always positive. The variation reflects the interaction between fault severity, substrate dynamics, and body dynamics: severe faults (knee freezes) produce large, immediate perturbations that the detector fires on quickly, while subtle faults (hip freeze) produce smaller perturbations that take longer to accumulate above the 3σ threshold.

The lead time is clinically relevant: in a real deployment, the lead time determines the window available for an engineered recovery policy to act. Our results show that even the shortest lead time (0.41 s for R-knee on Izhikevich) provides enough time for a simple recovery response (e.g., slowing down or stopping), while the longest lead times (up to 9 s, for the hip freeze) provide substantial opportunity for more sophisticated recovery strategies.

V. DISCUSSION

A. Seed robustness

Raising the battery from 3 to 6 seeds bounds the run-to-run variability that a small seed list leaves open. The AUC spread is honest: no seed falls below 0.68, and the 3σ detector threshold fires within 0.5 s of injection in every case. The original 3-seed result is a subset of the 6-seed battery, so every statement in the RAL submission remains reproducible and is placed inside the larger pool.

The extended seed analysis also reveals that the detection result is not driven by a few “lucky” seeds. The worst-case performance (AUC 0.683 for L-hip in seed 1) still represents a detector that performs substantially better than chance, and the mean performance (AUC 0.938) is robust across seeds. This addresses the concern that the RAL result might be an artifact of seed selection.

B. Substrate transfer

The Izhikevich substrate shows slightly lower AUC (mean $\Delta = -0.016$), but the fault signature transfers: both substrates detect every fault within 1.0 s of injection, the topographic localization is preserved for the distinct joint electrodes, and the dead-null control collapses every metric to chance. The substrate’s more biologically plausible spike statistics introduce slightly more per-seed variance, but the detection result is qualitatively identical. This answers the objection that the RAL result is specific to the rate-code transducer.

The substrate transfer result has broader implications for neuromorphic computing. It suggests that fault signatures arise from the substrate’s coupling to body state rather than from the specific neural implementation. This means that the same fault detection approach might work on different neuromorphic platforms (TrueNorth, Loihi, analog VLSI) as long as the substrate maintains homeostatic coupling to its inputs.

C. Terrain generalization

Detection remains robust across terrains: AUC is above 0.93 for every fault on every terrain. Lead time stays positive on all terrains, and detection still occurs well before collapse. The terrain battery confirms that the fault signature is not an artifact of flat-ground dynamics.

The terrain result is particularly important for real-world deployment, where walking conditions are rarely perfectly flat. The fact that the signature survives on rough terrain suggests that culture substrates could provide fault information in practical applications, even when the walking conditions vary.

D. Honesty: where the extensions do not help

The extensions do not change the authority result: a detector-triggered soft halt leaves survival unchanged across all substrates and terrains. The culture reports and localizes; it does not compensate. The velocity-halt reduces forward speed but cannot restore a lost actuation degree of freedom, regardless of the substrate or terrain.

This honesty is crucial for the practical value of the work. While the culture substrate provides valuable fault information, it cannot replace engineered recovery systems. The value of the culture is as a sensor that provides early warning, not as a controller that can compensate for faults.

E. Relation to model-based fault detection

Classical FDI treats the plant as a known dynamical system: residuals are generated from a nominal model and thresholded to raise an alarm [21]–[23]. Our detector deliberately uses *none* of that: the plant model is never invoked, no observer is built, and the only signal is the per-tick spike count of a substrate that was never trained on faults. In FDI terms the z -score alarm is a scalar residual taken on the

culture readout rather than on the plant, and the per-channel map plays the role of a structured residual vector [22].

The model-free nature of our approach is both a strength and a weakness. Strength: it does not require an analytical model of the plant, which may be unavailable or too complex to derive. Weakness: it cannot provide the detailed diagnostic information that model-based methods offer (e.g., isolating the specific fault mode, estimating fault severity). Our approach is complementary to model-based FDI, providing a first-stage detector that triggers more sophisticated analysis.

F. Implications for embodied neuromorphic systems

The measurement contract assumed here—sixty-four electrodes, per-tick encoding, a fixed hand-specified readout—is directly realizable on existing neuromorphic and high-density culture hardware [16], [18], [19]. The guarantee required of a deployed substrate is not that it *learns* the fault but that healthy statistics are stable enough for a fixed threshold to separate signal from noise—our seed-matched baseline [20]—so “report and localize” does not depend on fault-condition data.

This has practical implications for the design of neuromorphic control systems. If fault signatures are detectable from the spike stream, then neuromorphic substrates can serve dual purposes: providing control signals and monitoring for faults. This could reduce the need for separate sensing and monitoring hardware, simplifying system design and reducing power consumption.

G. Limitations

Three limits bound these claims. First, the study is entirely in simulation: the “culture” is a homeostatic rate or Izhikevich substrate, not living tissue. Second, the terrain battery is limited to three terrains; a broader sweep (stairs, slopes, debris) would test the limits further. Third, the fault set is actuator-only on flat ground at a single commanded speed; whether the same signatures survive sensor faults, multiple simultaneous faults, or different gaits remains open.

Additional limitations include:

- **Single fault injection time:** all faults are injected at $t = 2.0$ s. Whether the detector works equally well at different times (e.g., during startup or during different phases of the gait cycle) is untested.
- **Single walking speed:** the walker operates at $v_x \approx 0.5$ m/s. Whether the signature generalizes to different speeds (faster walking, running, standing) is unknown.
- **Single body configuration:** the study uses a single robot (Unitree G1) with a single set of joint parameters. Whether the signature transfers to different robots or different body configurations is untested.
- **No sensor faults:** the study focuses on actuator faults. Whether sensor faults (e.g., encoder failures, IMU drift) produce detectable signatures is an open question.
- **No multiple simultaneous faults:** the study injects one fault at a time. Whether the detector can handle multiple

simultaneous faults, or whether the signatures interfere, is unknown.

These limitations are all addressable with the same experimental framework, and we leave them to future work.

H. Future work

Several directions extend this work:

- **Physical validation:** the most important next step is to run the same protocol on a physical culture substrate (e.g., DishBrain, CL1) to determine whether the simulation results transfer to living tissue.
- **Additional fault types:** sensor faults, multiple simultaneous faults, and gradual degradation (aging, wear) would test the limits of the detection approach.
- **Additional terrains and gaits:** stairs, slopes, debris, and different walking speeds would test the generalization of the terrain result.
- **Model-based recovery:** combining the culture's early warning with model-based recovery controllers (e.g., capture-point stepping, whole-body optimization) could demonstrate practical value.
- **Online adaptation:** learning the detector threshold online (rather than using a fixed seed-matched baseline) could improve robustness to drift and changing conditions.

VI. COMPARISON WITH STATE OF THE ART

Table VI compares our approach with representative fault-detection and anomaly-detection methods for humanoid robots. The comparison is organized along five dimensions: detection performance (AUC), lead time, fault coverage, closed-loop operation, and whether the method requires a biological or neuromorphic substrate.

Three observations emerge from this comparison.

First, our detection performance is comparable to or better than proprioception-only methods. RAPT achieves 0.92 AUROC on general OOD detection using a GRU on proprioceptive data alone, while our approach reaches 0.767–0.997 on specific actuator faults using only the spike stream. The lead times are longer for our method (0.41–9.04 s vs. ~ 0.02 s), but this reflects the different detection paradigms: RAPT operates at 50 Hz control rate and flags deviations per tick, while our detector operates on the spike stream and flags when the z -score crosses a threshold. The longer lead time is actually an advantage for practical deployment, as it provides more time for an engineered recovery policy to act.

Second, our method is the only one that operates on a culture substrate. Classical FDI approaches [22], [24] require an explicit plant model; RAPT requires proprioceptive data and a nominal trajectory model; our method requires neither—the culture substrate naturally generates residuals through its response to changing input statistics. This makes our approach potentially applicable to systems where analytical models are unavailable or too complex to

derive, and where the culture substrate is already present for control purposes.

Third, the dead-null control is unique to our approach. None of the compared methods includes a substrate-matched null control that distinguishes between substrate-carried information and readout-channel artifacts. The matched dead Poisson substrate (same spike rate, no state coupling) collapses every metric to chance, confirming that the fault signature is carried by the substrate's coupling to body state. This falsification apparatus is essential for any substrate-based claim and is not present in the neuromorphic or BCI literature.

The comparison also highlights the complementary nature of our approach. RAPT and similar methods detect out-of-distribution behavior in the control signal; our method detects actuator faults in the spike stream. A practical system could combine both: RAPT monitors the control output while the culture substrate monitors the internal state, providing redundant fault coverage through independent sensing modalities.

VII. CONCLUSION

We extended the fault-detection study in three directions: seed robustness (3 to 6 seeds), substrate transfer (rate vs. Izhikevich), and terrain generalization (flat, slope, rough). The fault signature is robust: mean AUC 0.767–0.997 per fault across all conditions, detection well before collapse in every case, and topographic localization preserved for the distinct joint electrodes. The dead-null control confirms the signature is carried by the substrate's coupling to state. The culture is an informing sensor whose fault signatures transfer across substrates, seeds, and terrains, but whose authority to compensate remains bounded.

The extended analysis strengthens the original RAL result in several ways. First, the 6-seed battery demonstrates that the detection result is not seed-dependent: no seed falls below AUC 0.68, and the mean performance is robust (AUC 0.938). Second, the substrate transfer result shows that the signature is not specific to the rate-code transducer: the Izhikevich substrate, with its more biologically plausible dynamics, produces qualitatively identical results. Third, the terrain battery confirms that the signature is not an artifact of flat-ground dynamics: detection remains robust (AUC ≥ 0.93) even on rough terrain.

These results have practical implications for neuromorphic control systems. If fault signatures are detectable from the spike stream, then culture substrates can serve dual purposes: providing control signals and monitoring for faults. This could reduce the need for separate sensing and monitoring hardware, simplifying system design and reducing power consumption.

The limitation remains that the culture cannot compensate for faults it detects. The detector-triggered soft halt changes nothing in closed loop (0/3 per arm per fault), and only the variability fault, which leaves actuation intact, passes through. The culture belongs on the monitor, not the controller: it earns its place by announcing and localizing a fault

TABLE VI

COMPARISON WITH STATE-OF-THE-ART FAULT-DETECTION AND ANOMALY-DETECTION METHODS FOR HUMANOID ROBOTS. AUC IS REPORTED WHERE AVAILABLE; LEAD TIME IS THE INTERVAL BETWEEN DETECTION AND PHYSICAL COLLAPSE OR THRESHOLD CROSSING; “CL” DENOTES CLOSED-LOOP OPERATION; “SUBSTRATE” INDICATES WHETHER THE METHOD REQUIRES A BIOLOGICAL OR NEUROMORPHIC SUBSTRATE.

Method	Approach	Year	AUC Acc.	/	Lead time	Faults	CL	Substrate
Ours	z-score on spike stream	2026	0.767–0.997		0.41–9.04 s	4 actuator	Yes	Rate + Izh
RAPT [35]	GRU probabilistic trajectory model	2026	0.92 AU-ROC		~0.02 s	OOD (general)	Yes	None
Mungai [36]	1D CNN on proprioception	2023	1.0 FPR)	(0	1.6 s	Abrupt/independent	Yes	None
Isermann [22]	Model-based residual	2005	N/A		N/A	General	Yes	None
DishBrain drug [37]	Pharmacological modulation	2025	N/A		N/A	Drug response	Yes	Biological
De Luca [24]	Generalized momenta	2003	N/A		N/A	Actuator	Yes	None

*Mungai & Grizzle evaluate standing only; no dynamic walking task. RAPT uses Unitree G1 with proprioceptive input (no culture substrate).

early enough for an engineered recovery policy to act, and it should not be asked to close the loop it has just diagnosed.

Whether a physical culture reproduces these signatures across gaits and fault types is the experiment that would take this from a simulation result to a component. The framework presented here—fault injection, spike-stream analysis, dead-null control, and authority bounding—provides a rigorous template for that validation.

ACKNOWLEDGMENT

The author thanks the open-source robotics and neuromorphic communities whose tooling made this study possible, and the companion ICRA 2027 paper for the load-bearing protocol this fault battery builds on.

REFERENCES

- [1] B. J. Kagan *et al.*, “In vitro neurons learn and exhibit sentence when embodied in a simulated game-world,” *Neuron*, vol. 110, no. 23, pp. 3952–3969, 2022.
- [2] F. Balci, S. Ben Hamed, T. Boraud, S. Bouret, T. Brochier, C. Brun, *et al.*, “A response to claims of emergent intelligence and sentience in a dish,” *Neuron*, vol. 111, no. 5, pp. 604–605, 2023.
- [3] B. J. Kagan *et al.*, “Scientific communication and the semantics of sentence,” *Neuron*, vol. 111, no. 5, 2023. [letter]
- [4] W. Maass, “Networks of spiking neurons: the third generation of neural network models,” *Neural Networks*, vol. 10, no. 9, pp. 1659–1671, 1997.
- [5] E. M. Izhikevich, “Simple model of spiking neurons,” *IEEE Trans. Neural Networks*, vol. 14, no. 6, pp. 1569–1572, 2003.
- [6] E. M. Izhikevich, “Which model to use for cortical spiking neurons?” *IEEE Trans. Neural Networks*, vol. 15, no. 5, pp. 1063–1070, 2004.
- [7] W. Maass, T. Natschlaeger, and H. Markram, “Real-time computing without stable states: a new framework for neural computation based on perturbations,” *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, 2002.
- [8] C. Eliasmith and C. H. Anderson, *Neural Engineering: Computation, Representation, and Dynamics in Neurobiological Systems*. MIT Press, 2003.
- [9] T. Bekolay *et al.*, “Nengo: a Python tool for building large-scale functional brain models,” *Frontiers in Neuroinformatics*, vol. 7, art. 48, 2014.
- [10] T. Schreiber, “Measuring information transfer,” *Phys. Rev. Lett.*, vol. 85, no. 2, pp. 461–464, 2000.
- [11] A. Kraskov, H. Stögbauer, and P. Grassberger, “Estimating mutual information,” *Phys. Rev. E*, vol. 69, 066138, 2004.
- [12] S. Panzeri, R. Senatore, M. A. Montemurro, and R. S. Petersen, “Correcting for the sampling bias in spike-train information measures,” *J. Neurophysiology*, vol. 98, pp. 1064–1072, 2007.
- [13] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, and J. D. Farmer, “Testing for nonlinearity in time series: the method of surrogate data,” *Physica D*, vol. 58, pp. 77–94, 1992.
- [14] M. D. Ernst, “Permutation methods: a basis for exact inference,” *Statistical Science*, vol. 19, no. 4, pp. 676–685, 2004.
- [15] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: a physics engine for model-based control,” in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.
- [16] H. Cai *et al.*, “Brain organoid reservoir computing for artificial intelligence,” *Nature Electronics*, vol. 6, no. 12, pp. 1032–1039, 2023.
- [17] D. Verstraeten, B. Schrauwen, M. D’Haene, and D. Stroobandt, “An experimental unification of reservoir computing methods,” *Neural Networks*, vol. 20, no. 3, pp. 391–403, 2007.
- [18] P. A. Merolla *et al.*, “A million spiking-neuron integrated circuit with a scalable communication network and interface,” *Science*, vol. 345, no. 6197, pp. 668–673, 2014.
- [19] M. Davies *et al.*, “Loihi: a neuromorphic manycore processor with on-chip learning,” *IEEE Micro*, vol. 38, no. 1, pp. 82–99, 2018.
- [20] G. G. Turrigiano and S. B. Nelson, “Homeostatic plasticity in the

- developing nervous system,” *Nature Reviews Neuroscience*, vol. 5, no. 2, pp. 97–107, 2004.
- [21] V. Venkatasubramanian, R. Rengaswamy, K. Yin, and S. N. Kavuri, “A review of process fault detection and diagnosis, Part I: quantitative model-based methods,” *Computers & Chemical Engineering*, vol. 27, no. 3, pp. 293–311, 2003.
- [22] R. Isermann, “Model-based fault-detection and diagnosis — status and applications,” *Annual Reviews in Control*, vol. 29, no. 1, pp. 71–85, 2005.
- [23] J. Chen and R. J. Patton, *Robust Model-Based Fault Diagnosis for Dynamic Systems*. Boston, MA: Kluwer Academic Publishers, 1999.
- [24] A. De Luca and R. Mattone, “Actuator failure detection and isolation using generalized momenta,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2003, pp. 634–639.
- [25] M. L. Visinsky, J. R. Cavallaro, and I. D. Walker, “A dynamic fault tolerance framework for remote robots,” *IEEE Trans. Robotics and Automation*, vol. 11, no. 4, pp. 477–490, 1995.
- [26] A. P. Georgopoulos, A. B. Schwartz, and R. E. Kettner, “Neuronal population coding of movement direction,” *Science*, vol. 233, no. 4771, pp. 1416–1419, 1986.
- [27] S. Kajita *et al.*, “Biped walking pattern generation by using preview control of zero-moment point,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2003, pp. 1620–1626.
- [28] J. Pratt, J. Carff, S. Drakunov, and A. Goswami, “Capture point: a step toward humanoid push recovery,” in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2006, pp. 200–207.
- [29] P.-B. Wieber, “Trajectory free linear model predictive control for stable walking in the presence of strong perturbations,” in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2006, pp. 137–142.
- [30] S. Kuindersma *et al.*, “Optimization-based locomotion planning, control, and simulation studies with the Atlas humanoid robot,” *Autonomous Robots*, vol. 40, no. 3, pp. 429–455, 2016.
- [31] J. Hwangbo *et al.*, “Learning agile and dynamic motor skills for legged robots,” *Science Robotics*, vol. 4, no. 26, 2019.
- [32] M. Kalakrishnan, J. Buchli, P. Pastor, M. Mistry, and S. Schaal, “Fast, robust quadruped locomotion over unknown terrain,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2010, pp. 4672–4677.
- [33] R. Grandia, F. Farshidian, R. Ranjan, and M. Hutter, “Hierarchical MPC for robust multi-gait locomotion of legged robots,” *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1471–1478, 2021.
- [34] P. Fankhauser *et al.*, “Robust fast navigation of autonomous mobile robots in natural environments,” *Autonomous Robots*, vol. 42, no. 7, pp. 1435–1452, 2018.
- [35] H. Munn, B. Tidd, P. Böhm, M. Gallagher, and D. Howard, “RAPT: Model-predictive out-of-distribution detection and failure diagnosis for sim-to-real humanoid deployment,” *arXiv preprint arXiv:2602.01515*, 2026.
- [36] M. E. Mungai, G. Prabhakaran, and J. W. Grizzle, “Fall prediction for bipedal robots: the standing phase,” *arXiv preprint arXiv:2309.14546*, 2023.
- [37] B. J. Kagan *et al.*, “Drug treatment alters performance in a neural microphysiological system of information processing,” *Communications Biology*, vol. 8, 2025.